

Yuning You

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RESEARCH FOCUS

Machine learning algorithms for structural data mining (e.g. [graphs](#), [point clouds](#), and [fields](#)), and their use in building biological foundation models to simulate living organisms at multiple scales (e.g. [virtual tissues](#)).

EXPERIENCES

-  **The Chinese University of Hong Kong, Shenzhen**, Guangdong Sep 2025 – **Present**
Assistant Professor in School of Science and Engineering
Affiliated with Shenzhen Future Network of Intelligence Institute
Affiliated with Shenzhen Key Laboratory of Big Data, and Artificial Intelligence
-  **California Institute of Technology**, Pasadena, California Jul 2024 – Aug 2025
Postdoctoral Scholar in Division of Biology and Biological Engineering
Advisor: Prof. [Matt Thomson](#)
Guest Collaborator Aug 2025 – **Present**
-  **Texas A&M University**, College Station, Texas Aug 2019 – Aug 2024
Ph.D. in Electrical Engineering
Graduate Research Assistant in Department of Electrical and Computer Engineering (Part-Time)
Sep 2020 – May 2024
Advisors: Prof. [Yang Shen](#) & Prof. [Zhangyang \(Atlas\) Wang](#)
-  **Xi'an Jiaotong University**, Xi'an, Shaanxi Aug 2015 – Jun 2019
B.Eng. in Information Engineering
-  **Genentech, Inc.**, South San Francisco, California May 2023 – Aug 2023
Early Clinical Development/AI/ML Intern in Genentech Research and Early Development
-  **insitro, Inc.**, South San Francisco, California May 2022 – Aug 2022
ML Small Molecules Intern in Department of Data Science and Machine Learning
-  **Amazon.com Services, Inc.**, Remote Jun 2021 – Aug 2021
Applied Scientist Intern in Product Semantics Team

PUBLICATIONS (Please check the full list on [\[Google Scholar\]](#))

- LoG'25** [\[link\]](#): “When Structure Doesn’t Help: LLMs Do Not Read Text-Attributed Graphs as Effectively as We Expected”, H. Xu, **Y. You**, T. Ma, *Learning on Graphs Conference*, 2025.
- MLGenX'25** [\[link\]](#): “Building Foundation Models to Characterize Cellular Interactions via Geometric Self-Supervised Learning on Spatial Genomics”, **Y. You**, Z. Wang, K. Fleisher, R. Liu, M. Thomson, *Machine Learning for Genomics Explorations Workshop, International Conference on Learning Representations*, 2025.
- AIDrugX@NeurIPS'24** [\[link\]](#): “Correlational Lagrangian Schrödinger Bridge: Learning Dynamics with Population-Level Regularization”, **Y. You**, R. Zhou, Y. Shen, *AI for New Drug Modalities Workshop, Conference on Neural Information Processing Systems*, 2024.
- HUGO'24** [\[link\]](#): “Critical Assessment of Variant Prioritization Methods for Rare Disease Diagnosis within the Rare Genomes Project”, ..., **Y. You**, ..., *Human Genomics*, vol. 18(44), 2024. (Impact Factor 4.50, Outcome of [CAGI6 RGP](#))
- ICLR'24** [\[link\]](#): “Latent 3D Graph Diffusion”, **Y. You**, R. Zhou, J. Park, H. Xu, C. Tian, Z. Wang, Y. Shen, *International Conference on Learning Representations*, oprev., 2024. (Acceptance Rate 31.00%)

NeurIPS'23 [\[link\]](#): “Graph Mixture of Experts: Learning on Large-Scale Graphs with Explicit Diversity Modeling”, H. Wang, Z. Jiang, **Y. You**, Y. Han, G. Liu, J. Srinivasa, R. Kompella, Z. Wang, *Conference on Neural Information Processing Systems*, pp. 50825-50837, 2023. (Acceptance Rate 26.10%)

ICLR'23 [\[link\]](#): “Graph Domain Adaptation via Theory-Grounded Spectral Regularization”, **Y. You**, T. Chen, Z. Wang, Y. Shen, *International Conference on Learning Representations*, oprev., 2023. (Acceptance Rate 31.80%)

NeurIPS'22 [\[link\]](#): “Augmentations in Hypergraph Contrastive Learning: Fabricated and Generative”, T. Wei*, **Y. You***, T. Chen, Y. Shen, J. He, Z. Wang, *Conference on Neural Information Processing Systems*, pp. 1909-1922, 2022. (*Equal Contribution, Acceptance Rate 25.60%)

Bioinformatics'22 [\[link\]](#): “Cross-Modality and Self-Supervised Protein Embedding for Compound-Protein Affinity and Contact Prediction”, **Y. You**, Y. Shen, *Bioinformatics*, vol. 38(Supplement_2), pp. 68-74, 2022. (Impact Factor 6.93, MoML'22, ECCB'22 with Acceptance Rate 17.40%, 3DSIG COSI@ISMB/ECCB'21, MLSB@NeurIPS'20)

ICLR'22 [\[link\]](#): “Bayesian Modeling and Uncertainty Quantification for Learning to Optimize: What, Why, and How”, **Y. You**, Y. Cao, T. Chen, Z. Wang, Y. Shen, *International Conference on Learning Representations*, oprev., 2022. (Acceptance Rate 32.29%)

WSDM'22 [\[link\]](#): “Bringing Your Own View: Graph Contrastive Learning without Prefabricated Data Augmentations”, **Y. You**, T. Chen, Z. Wang, Y. Shen, *ACM International Conference on Web Search and Data Mining*, pp. 1300-1309, 2022. (Acceptance Rate 20.22%)

ICML'21 Long Presentation [\[link\]](#): “Graph Contrastive Learning Automated”, **Y. You**, T. Chen, Y. Shen, Z. Wang, *International Conference on Machine Learning*, pp. 12121-12132, 2021. (Acceptance Rate 3.01%)

TVT'21 [\[link\]](#): “Probabilistic Constructive Interference Precoding for Imperfect CSIT”, G. Lyu, **Y. You**, A. Li, X. Liao, C. Masouros, *IEEE Transactions on Vehicular Technology*, vol. 70(4), pp. 3932-3937, 2021. (Impact Factor 5.97)

NeurIPS'20 [\[link\]](#): “Graph Contrastive Learning with Augmentations”, **Y. You***, T. Chen*, Y. Sui, T. Chen, Z. Wang, Y. Shen, *Conference on Neural Information Processing Systems*, pp. 5812-5823, 2020. (*Equal Contribution, Acceptance Rate 20.09%)

ICML'20 [\[link\]](#): “When Does Self-Supervision Helps Graph Convolutional Networks?”, **Y. You***, T. Chen*, Z. Wang, Y. Shen, *International Conference on Machine Learning*, pp. 10871-10880, 2020. (*Equal Contribution, Acceptance Rate 21.80%)

CVPR'20 [\[link\]](#): “L²-GCN: Layer-Wise and Learned Efficient Training of Graph Convolutional Networks”, **Y. You***, T. Chen*, Z. Wang, Y. Shen, *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 2127-2135, 2020. (*Equal Contribution, Acceptance Rate 22.08%)

AWARDS

Presidential Young Fellow, The Chinese University of Hong Kong, Shenzhen	Nov 2025
Top 2% Scientists in 2024, Departments of METRICS, Stanford University/Elsevier [link]	Sep 2025
Distinguished Graduate Student Award for Excellence in Research, Association of Former Students of Texas A&M University (9 awardees) [link]	Mar 2025
ECEN Quality Graduate Student Award, Department of Electrical and Computer Engineering, Texas A&M University	Apr 2023
NSF Student Travel Awards, ACM International Conference on Web Search and Data Mining	Dec 2021

Chevron Scholarship, Department of Electrical and Computer Engineering, Texas A&M University
Sep 2021

Electrical and Computer Engineering PhD Merit Fellowship, Department of Electrical and Computer Engineering, Texas A&M University
Feb 2019

TALKS

SSE Weekly Colloquium, The Chinese University of Hong Kong, Shenzhen [link]	Mar 2026
CIoTSC'25, Multimodal World Models for Embodied Robotics and Autonomous Driving Workshop, Suzhou [link]	Oct 2025
University of Southern California, Prof. Leonardo Morsut 's Lab, Los Angeles	Aug 2025
Texas A&M University, Prof. James Cai 's Lab, online	Oct 2023
Genentech, Inc., Spatial Omics Journal Club, online	Aug 2023
AstraZeneca plc, AI&A Journal Club, online	Mar 2022
University of Texas at Austin, Prof. Mingyuan Zhou 's Group, online	Oct 2021
Technical University of Munich, Learning on Graphs and Geometry Reading Group (LoGaG), online	Aug 2021
ISMB/ECCB'21, 3DSIG COSI: Structural Bioinformatics and Computational Biophysics , online [video]	Jul 2021
ICML'21, Session of Semisupervised and Unsupervised Learning , online [video]	Jul 2021

SERVICES

Area Chair of [New Perspectives in Advancing Graph Machine Learning Workshop](#) at NeurIPS'25

Co-Organizer of [AI Bootcamp VIII on Graph Machine Learning](#) at Caltech

Session Chair of [Semisupervised and Unsupervised Learning](#) at ICML'21

Reviewer in Conferences of ICML'21-24, NeurIPS'21-24, ICLR'22,24-25, WWW'22, LoG'22,24, ISMB/ECCB'21,23-24, ACM-BCB'21,23,24

Reviewer in Journals of TPAMI'21,23, TMLR'23, TNNLS'21-23, TKDE'22, TAI'22, INS'21, PeerJ'21, NEPL'21, JCST'22,24, SIPN'22, INFFUS'23, JBS'23, CSUR'24, Nature Communications Chemistry'25, Nature Computational Science'25, Journal of the American Statistical Association'26